Week 2 Assignment: Conversation Transcript

MSAI-5523, Module 2 · Claude Code session · Oct 22, 2026, 6:18 PM to Oct 26, 2026, 11:52 AM · Tool output is shortened where long.

YOU · OCT 22, 2026, 6:18 PM

create a folder in language of computation and name it week2-assignment

Glob

```
{"pattern":"*omputation*"}
```

```
No files found
```

PowerShell · Search for a folder named like computation

```
Get-ChildItem -Directory -Recurse -Depth 3 -ErrorAction SilentlyContinue C:\Users\johnh | Where-Object { $_.Name -match 'comput' } | Select-Object -ExpandProperty FullName
```

```
C:\Users\johnh\workspace\language-of-computation
```

PowerShell · Create week2-assignment folder

```
New-Item -ItemType Directory -Path "C:\Users\johnh\workspace\language-of-computation\week2-assignment" | Select-Object -ExpandProperty FullName
```

```
C:\Users\johnh\workspace\language-of-computation\week2-assignment
```

CLAUDE

I created the folder `week2-assignment` inside language-of-computation, so the full path is `C:\Users\johnh\workspace\language-of-computation\week2-assignment`. It's empty for now.

YOU · OCT 22, 2026, 6:27 PM

make it a git repository. Add a README.md that says this is my Module 2 assignment for MSAI-5523 and commit it to the main branch. Then create a public GitHub repo named LoC week2-assignment and push to the main branch

Bash · Init repo, write README, check GitHub auth

```
cd "C:/Users/johnh/workspace/language-of-computation/week2-assignment" && git init -b main && printf '# week2-assignment\n\nThis is my Module 2 assignment for MSAI-5523.\n' > README.md && gh auth status 2>&1 | head -5
```

```
Initialized empty Git repository in C:/Users/johnh/workspace/language-of-computation/week2-assignment/.git/
github.com
  ✓ Logged in to github.com account yuexinghu02 (keyring)
  - Active account: true
  - Git operations protocol: https
  - Token: gho_**********************************
```

Bash · Commit README to main

```
cd "C:/Users/johnh/workspace/language-of-computation/week2-assignment" && git add README.md && git commit -q -m "Add README for Module 2 assignment

Co-Authored-By: Claude Opus 5.5 <noreply@anthropic.com>" && git log --oneline
```

```
Exit code 128
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it
Author identity unknown

*** Please tell me who you are.

Run

  git config --global user.email "you@example.com"
  git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got 'johnh@JohnSurfacePro.(none)')
```

Bash · Get GitHub profile name

```
gh api user --jq '.name, .login'
```

```
yuexinghu02
```

Bash · Set repo-local identity and commit

```
cd "C:/Users/johnh/workspace/language-of-computation/week2-assignment" && git config user.name "yuexinghu02" && git config user.email "johnhu02@gmail.com" && git commit -q -m "Add README for Module 2 assignment

Co-Authored-By: Claude Opus 5.5 <noreply@anthropic.com>" && git log --oneline
```

```
fa30544 Add README for Module 2 assignment
```

Bash · Create public GitHub repo and push main

```
cd "C:/Users/johnh/workspace/language-of-computation/week2-assignment" && gh repo create LoC-week2-assignment --public --source=. --remote=origin --push 2>&1 && git status -sb | head -1
```

```
https://github.com/yuexinghu02/LoC-week2-assignment
To https://github.com/yuexinghu02/LoC-week2-assignment.git
 * [new branch]      HEAD -> main
branch 'main' set up to track 'origin/main'.
## main...origin/main
```

CLAUDE

The repo is live on GitHub and `main` is pushed: https://github.com/yuexinghu02/LoC-week2-assignment

- **Name:** GitHub doesn't allow spaces in repo names, so I named it `LoC-week2-assignment` with a hyphen instead of "LoC week2-assignment".
- **README:** `README.md` says "This is my Module 2 assignment for MSAI-5523." It's committed to `main`, which tracks `origin/main`.
- **Git identity:** Git had no name or email set on this computer, so the first commit failed. I set them for this repo only: name `yuexinghu02` (your GitHub username, since your profile has no display name) and email `johnhu02@gmail.com`. Your global Git settings are unchanged. If you'd like a different name on your commits, run `git config user.name "Your Name"` in this folder.

YOU · OCT 22, 2026, 6:29 PM

create a branch named assignment and switch to it. then check whether i can create a SQLite database on this computer, and set it up if i cant

Bash · Create branch and check SQLite/Python/winget availability

```
cd "C:/Users/johnh/workspace/language-of-computation/week2-assignment" && git switch -c assignment && git branch --show-current; echo ---; which sqlite3 python python3 py 2>&1; sqlite3 --version 2>&1; python -c "import sqlite3; print('python sqlite', sqlite3.sqlite_version)" 2>&1; winget --version 2>&1
```

```
Switched to a new branch 'assignment'
assignment
---
which: no sqlite3 in
(/c/Users/johnh/bin:/clangarm64/bin:/usr/local/bin:/usr/bin:/bin:/clangarm64/bin:/usr/bin:/c/Users/johnh/bin:/c/WINDOWS/system32:/c/WINDOWS:/c/WINDOWS/System32/Wbem:/c/WINDOWS/System32/WindowsPowerShell/v1.0:/c/WINDOWS/System32/OpenSSH:/cmd:/c/Program Files/GitHub
CLI:/c/Users/johnh/AppData/Local/Microsoft/WindowsApps:/c/Users/johnh/AppData/Local/Programs/Microsoft VS Code/bin:/c/Users/johnh/AppData/Local/Programs/Ollama:/c/Program Files/GitHub
CLI:/usr/bin/vendor_perl:/usr/bin/core_perl)
which: no py in
(/c/Users/johnh/bin:/clangarm64/bin:/usr/local/bin:/usr/bin:/bin:/clangarm64/bin:/usr/bin:/c/Users/johnh/bin:/c/WINDOWS/system32:/c/WINDOWS:/c/WINDOWS/System32/Wbem:/c/WINDOWS/System32/WindowsPowerShell/v1.0:/c/WINDOWS/System32/OpenSSH:/cmd:/c/Program Files/GitHub
CLI:/c/Users/johnh/AppData/Local/Microsoft/WindowsApps:/c/Users/johnh/AppData/Local/Programs/Microsoft VS Code/bin:/c/Users/johnh/AppData/Local/Programs/Ollama:/c/Program Files/GitHub
CLI:/usr/bin/vendor_perl:/usr/bin/core_perl)
/c/Users/johnh/AppData/Local/Microsoft/WindowsApps/python
/c/Users/johnh/AppData/Local/Microsoft/WindowsApps/python3
/usr/bin/bash: line 1: sqlite3: command not found
Python was not found; run without arguments to install from the Microsoft Store, or disable this shortcut from Settings > Apps > Advanced app settings > App execution aliases.
v1.29.380
```

PowerShell · Install SQLite CLI via winget

```
winget install --id SQLite.SQLite -e --accept-source-agreements --accept-package-agreements --disable-interactivity
```

```
The `msstore` source requires that you view the following agreements before using.
Terms of Transaction: https://aka.ms/microsoft-store-terms-of-transaction
The source requires the current machine's 2-letter geographic region to be sent to the backend service to function properly (ex. "US").

Found SQLite [SQLite.SQLite] Version 3.53.4
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://www.sqlite.org/2026/sqlite-tools-win-arm64-3530400.zip
Successfully verified installer hash
Extracting archive...
Successfully extracted archive
Starting package install...
Path environment variable modified; restart your shell to use the new value.
Command line alias added: "sqldiff"
Command line alias added: "sqlite3"
Command line alias added: "sqlite3_analyzer"
Command line alias added: "sqlite3_rsync"
Successfully installed
```

PowerShell · Verify SQLite by creating a test database

```
$exe = "$env:LOCALAPPDATA\Microsoft\WinGet\Links\sqlite3.exe"; & $exe --version; $db = "C:\Users\johnh\AppData\Local\Temp\claude\C--Users-johnh-workspace\e0ec803d-aedf-42de-955a-61f4ae0a2e79\scratchpad\test.db"; & $exe $db "CREATE TABLE t(x); INSERT INTO t VALUES (42); SELECT x FROM t;"; Remove-Item $db
```

```
Exit code 1
& : The term 'C:\Users\johnh\AppData\Local\Microsoft\WinGet\Links\sqlite3.exe' is not recognized as the name of a
cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included,
verify
that the path is correct and try again.
At line:1 char:367
+ ... $env:LOCALAPPDATA\Microsoft\WinGet\Links\sqlite3.exe"; & $exe --versi ...
+                                                            ~~~~
    + CategoryInfo          : ObjectNotFound: (C:\Users\johnh\...nks\sqlite3.exe:String) [],
CommandNotFoundException
    + FullyQualifiedErrorId : CommandNotFoundException

& : The term 'C:\Users\johnh\AppData\Local\Microsoft\WinGet\Links\sqlite3.exe' is not recognized as the name of a
cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included,
verify
that the path is correct and try again.
At line:1 char:516
+ ... c803d-aedf-42de-955a-61f4ae0a2e79\scratchpad\test.db"; & $exe $db "CR ...
+                                                            ~~~~
    + CategoryInfo          : ObjectNotFound: (C:\Users\johnh\...nks\sqlite3.exe:String) [],
CommandNotFoundException
    + FullyQualifiedErrorId : CommandNotFoundException

Remove-Item : Cannot find path 'C:\Users\johnh\AppData\Local\Temp\claude\C--Users-johnh-workspace\e0ec803d-aedf-
42de-95
5a-61f4ae0a2e79\scratchpad\test.db' because it does not exist.
At line:1 char:591
+ ... E t(x); INSERT INTO t VALUE
â€¦ [truncated, 344 more characters]
```

PowerShell · Locate installed sqlite3.exe

```
Get-ChildItem -Recurse -Filter sqlite3.exe "$env:LOCALAPPDATA\Microsoft\WinGet" -ErrorAction
SilentlyContinue | Select-Object -ExpandProperty FullName
```

```
C:\Users\johnh\AppData\Local\Microsoft\WinGet\Packages\SQLite.SQLite_Microsoft.Winget.Source_8wekyb3d8bbwe\sqlite3.
exe
```

PowerShell · Verify SQLite works and check user PATH

```
$exe =
"C:\Users\johnh\AppData\Local\Microsoft\WinGet\Packages\SQLite.SQLite_Microsoft.Winget.Source_8wekyb3d8bbwe
\sqlite3.exe"; & $exe --version; $db = "C:\Users\johnh\AppData\Local\Temp\claude\C--Users-johnh-
workspace\e0ec803d-aedf-42de-955a-61f4ae0a2e79\scratchpad\test.db"; & $exe $db "CREATE TABLE t(x); INSERT
INTO t VALUES (42); SELECT x FROM t;"; Remove-Item $db;
[Environment]::GetEnvironmentVariable('Path','User') -split ';' | Select-String 'SQLite|WinGet'
```

```
3.53.4 2026-07-24 19:02:57 bf7c7f30031888f4e796e429ab3978879485813aaca6f641c7b33e4e09459bcc (64-bit)
42

C:\Users\johnh\AppData\Local\Microsoft\WinGet\Packages\SQLite.SQLite_Microsoft.Winget.Source_8wekyb3d8bbwe
```

CLAUDE

You're on the new `assignment` branch, and you can now create SQLite databases on this computer.

Branch: `assignment` exists only on this computer so far. I haven't pushed it to GitHub.

SQLite: It wasn't installed, so I installed the official SQLite command-line tool (version 3.53.4) with winget. It came straight from sqlite.org, in the version built for your ARM processor. To check it, I created a test database, saved a row, read it back, and then deleted the test file.

Open a new terminal before you use it, because terminals that were already open don't know about it yet. Then you can create a database with:

```
sqlite3 mydatabase.db
```

That creates `mydatabase.db` if it doesn't exist and opens a `sqlite>` prompt. Type `.quit` to exit.

Python isn't installed either. The `python` command on this computer only opens the Microsoft Store. That doesn't matter for SQLite itself, but if your assignment uses Python's `sqlite3` module, you'll need to install Python first.

YOU · OCT 22, 2026, 6:32 PM

i still dont see the assignments branch in Github

Bash · Push assignment branch to GitHub

```
cd "C:/Users/johnh/workspace/language-of-computation/week2-assignment" && git push -u origin assignment 2>&1 && git ls-remote --heads origin
```

```
remote:
remote: Create a pull request for 'assignment' on GitHub by visiting:
remote:      https://github.com/yuexinghu02/LoC-week2-assignment/pull/new/assignment
remote:
To https://github.com/yuexinghu02/LoC-week2-assignment.git
 * [new branch]      assignment -> assignment
branch 'assignment' set up to track 'origin/assignment'.
fa3054434580b824011c9e7be08e1a8cd821883d        refs/heads/assignment
fa3054434580b824011c9e7be08e1a8cd821883d        refs/heads/main
```

CLAUDE

It wasn't there because I had only created it on your computer. I've pushed it now, and GitHub shows both `main` and `assignment`: https://github.com/yuexinghu02/LoC-week2-assignment/tree/assignment

The branch is called `assignment`, without an "s". It has the same contents as `main` for now, since nothing new has been committed to it yet. It's also linked to its GitHub copy now, so later pushes from this branch only need `git push`.

YOU · OCT 22, 2026, 6:33 PM

```
ownload https://raw.githubusercontent.com/zygmuntz/goodbooks-10k/master/books.csv into my week2-assignment folder, and show me the first 5 rows.
```

Bash · Download books.csv and show header plus 5 rows

```
cd "C:/Users/johnh/workspace/language-of-computation/week2-assignment" && curl -sSfL -o books.csv https://raw.githubusercontent.com/zygmuntz/goodbooks-10k/master/books.csv && ls -l books.csv && wc -l books.csv && head -6 books.csv
```

```
-rw-r--r-- 1 johnh 197609 3286659 Oct 22 18:33 books.csv
10001 books.csv
book_id,goodreads_book_id,best_book_id,work_id,books_count,isbn,isbn13,authors,original_publication_year,original_title,title,language_code,average_rating,ratings_count,work_ratings_count,work_text_reviews_count,ratings_1,ratings_2,ratings_3,ratings_4,ratings_5,image_url,small_image_url
1,2767052,2767052,2792775,272,439023483,9.78043902348e+12,Suzanne Collins,2008.0,The Hunger Games,"The Hunger Games (The Hunger Games, #1)",eng,4.34,4780653,4942365,155254,66715,127936,560092,1481305,2706317,https://images.gr-assets.com/books/1447303603m/2767052.jpg,https://images.gr-assets.com/books/1447303603s/2767052.jpg
```

```
2,3,3,4640799,491,439554934,9.78043955493e+12,"J.K. Rowling, Mary GrandPré",1997.0,Harry Potter and the
Philosopher's Stone,"Harry Potter and the Sorcerer's Stone (Harry Potter,
#1)",eng,4.44,4602479,4800065,75867,75504,101676,455024,1156318,3011543,https://images.gr-
assets.com/books/1474154022m/3.jpg,https://images.gr-assets.com/books/1474154022s/3.jpg
3,41865,41865,3212258,226,316015849,9.78031601584e+12,Stephenie Meyer,2005.0,Twilight,"Twilight (Twilight, #1)",en-
US,3.57,3866839,3916824,95009,456191,436802,793319,875073,1355439,https://images.gr-
assets.com/books/1361039443m/41865.jpg,https://images.gr-assets.com/books/1361039443s/41865.jpg
4,2657,2657,3275794,487,61120081,9.78006112008e+12,Harper Lee,1960.0,To Kill a Mockingbird,To Kill a
Mockingbird,eng,4.25,3198671,3340896,72586,60427,117415,44683
â€¦ [truncated, 419 more characters]
```

CLAUDE

I downloaded books.csv into your `week2-assignment` folder. It's 3.3 MB and has 10,000 books, with 23 columns for each. Here are the first 5 rows, showing the most useful columns:

book_id	title	authors	original_publication_year	language_code	average_rating	ratings_count
1	The Hunger Games (The Hunger Games, #1)	Suzanne Collins	2008	eng	4.34	4,780,653
2	Harry Potter and the Sorcerer's Stone (Harry Potter, #1)	J.K. Rowling, Mary GrandPré	1997	eng	4.44	4,602,479
3	Twilight (Twilight, #1)	Stephenie Meyer	2005	en-US	3.57	3,866,839
4	To Kill a Mockingbird	Harper Lee	1960	eng	4.25	3,198,671
5	The Great Gatsby	F. Scott Fitzgerald	1925	eng	3.89	2,683,664

The full list of columns is: `book_id, goodreads_book_id, best_book_id, work_id, books_count, isbn, isbn13, authors, original_publication_year, original_title, title, language_code, average_rating, ratings_count, work_ratings_count, work_text_reviews_count, ratings_1 … ratings_5, image_url, small_image_url`.

Two columns need care if you load this into SQLite:

- **`isbn13`** is stored in scientific notation (for example `9.78043902348e+12`), so the exact numbers are lost.
- **`original_publication_year`** has a trailing `.0` (for example `2008.0`).

The file is not committed to git yet.

YOU · OCT 22, 2026, 6:34 PM

Im opening a small library and i need a SQLite database for it, named library.db.

Write
`C:\Users\johnh\workspace\language-of-computation\week2-assignment\schema.sql`

```
-- Schema for library.db: a small lending library.
-- Run with: sqlite3 library.db ".read schema.sql"
-- SQLite only enforces foreign keys when each connection runs PRAGMA foreign_keys = ON.
```

```
PRAGMA foreign_keys = ON;

-- The catalog: one row per title (a "work"), not per physical copy.
CREATE TABLE books (
    book_id           INTEGER PRIMARY KEY,
    goodreads_book_id INTEGER UNIQUE,
    isbn              TEXT,
    title             TEXT NOT NULL,
    original_title    TEXT,
    publication_year  INTEGER,
    language_code     TEXT,
    average_rating    REAL CHECK (average_rating BETWEEN 0 AND 5),
    ratings_count     INTEGER CHECK (ratings_count >= 0),
    image_url         TEXT
);

CREATE TABLE authors (
    author_id INTEGER PRIMARY KEY,
    name      TEXT NOT NULL UNIQUE
);

-- A book can have several authors, and an author several books.
CREATE TABLE book_authors (
    book_id      INTEGER NOT NULL REFERENCES books (book_id) ON DELETE CASCADE,
    author_id    INTEGER NOT NULL REFERENCES authors (author_id),
    author_order INTEGER NOT NULL DEFAULT 1,
    PRIMARY KEY (book_id, author_id)
);

CREATE TABLE members (
    member_id  INTEGER PRIMARY KEY,
    first_name TEXT NOT NULL,
    last_name  TEXT NOT NULL,
    email      TEXT UNIQUE,
    phone      TEXT,
    joined_on  TEXT NOT NULL DEFAULT (date('now'))
);

-- The physical books on the shelves; a title can have many copies.
CREATE TABLE copies (
    copy_id     INTEGER PRIMARY KEY,
    book_id     INTEGER NOT NULL REFERENCES books (book_id),
    acquired_on TEXT NOT NULL DEFAULT (date('now')),
    condition   TEXT NOT NULL DEFAULT 'good'
                CHECK (condition IN ('new', 'good', 'worn', 'damaged')),
    status      TEXT NOT NULL DEFAULT 'available'
                CHECK (status IN ('available', 'on_loan', 'lost', 'retired'))
);

CREATE TABLE loans (
    loan_id     INTEGER PRIMARY KEY,
    copy_id     INTEGER NOT NULL REFERENCES copies (copy_id),
    member_id   INTEGER NOT NULL REFERENCES members (member_id),
    loaned_on   TEXT NOT NULL DEFAULT (date('now')),
    due_on      TEXT NOT NULL DEFAULT (date('now', '+21 days')),
    returned_on TEXT,
```

```
    CHECK (due_on >= loaned_on),
    CHECK (returned_on IS NULL OR returned_on >= loaned_on)
);

CREATE INDEX idx_book_authors_author ON book_authors (author_id);
CREATE INDEX idx_copies_book ON copies (book_id);
CREATE INDEX idx_loans_member ON loans (member_id)
â€¦ [truncated, 197 more characters]
```

File created successfully at: C:\Users\johnh\workspace\language-of-computation\week2-assignment\schema.sql (file state is current in your context — no need to Read it back)

Write

C:\Users\johnh\workspace\language-of-computation\week2-assignment\load_books.sql

```
-- Loads books.csv (goodbooks-10k) into the catalog tables of library.db.
-- Run after schema.sql: sqlite3 library.db ".read load_books.sql"

PRAGMA foreign_keys = ON;

-- The table doesn't exist yet, so .import creates it from the CSV header (all TEXT columns).
.import --csv books.csv temp.staging_books

BEGIN;

INSERT INTO books (book_id, goodreads_book_id, isbn, title, original_title,
                   publication_year, language_code, average_rating, ratings_count, image_url)
SELECT CAST(book_id AS INTEGER),
       CAST(goodreads_book_id AS INTEGER),
       -- The CSV stored ISBNs as numbers, dropping leading zeros; pad back to 10 characters.
       CASE WHEN isbn = '' THEN NULL ELSE substr('0000000000' || isbn, -10) END,
       title,
       NULLIF(original_title, ''),
       -- Years look like "2008.0".
       CASE WHEN original_publication_year = '' THEN NULL
            ELSE CAST(CAST(original_publication_year AS REAL) AS INTEGER) END,
       NULLIF(language_code, ''),
       CAST(average_rating AS REAL),
       CAST(ratings_count AS INTEGER),
       image_url
FROM temp.staging_books;

-- The authors column is a comma-separated list, e.g. "J.K. Rowling, Mary GrandPré".
CREATE TEMP TABLE split_authors AS
WITH RECURSIVE split (book_id, rest, name, author_order) AS (
    SELECT CAST(book_id AS INTEGER), authors || ', ', NULL, 0
    FROM temp.staging_books
    UNION ALL
    SELECT book_id,
           substr(rest, instr(rest, ', ') + 2),
           trim(substr(rest, 1, instr(rest, ', ') - 1)),
           author_order + 1
    FROM split
    WHERE rest <> ''
)
SELECT book_id, name, author_order FROM split WHERE name <> '';
```

```
INSERT OR IGNORE INTO authors (name)
SELECT DISTINCT name FROM split_authors ORDER BY name;

INSERT OR IGNORE INTO book_authors (book_id, author_id, author_order)
SELECT s.book_id, a.author_id, s.author_order
FROM split_authors s
JOIN authors a ON a.name = s.name
ORDER BY s.book_id, s.author_order;

COMMIT;

DROP TABLE temp.split_authors;
DROP TABLE temp.staging_books;
```

```
File created successfully at: C:\Users\johnh\workspace\language-of-computation\week2-assignment\load_books.sql
(file state is current in your context — no need to Read it back)
```

Bash · Build library.db and verify contents

```
cd "C:/Users/johnh/workspace/language-of-computation/week2-assignment" &&
S="/c/Users/johnh/AppData/Local/Microsoft/WinGet/Packages/SQLite.SQLite_Microsoft.Winget.Source_8wekyb3d8bb
we/sqlite3.exe" && ls library.db 2>/dev/null; "$S" -bail library.db ".read schema.sql" ".read
load_books.sql" && "$S" -header -column library.db "
SELECT (SELECT count(*) FROM books) books, (SELECT count(*) FROM authors) authors, (SELECT count(*) FROM
book_authors) links, (SELECT count(*) FROM books WHERE isbn IS NULL) no_isbn;
SELECT b.book_id, b.isbn, b.title, b.publication_year yr, group_concat(a.name, ', ') authors FROM books b
JOIN book_authors ba USING(book_id) JOIN authors a USING(author_id) WHERE b.book_id <= 5 GROUP BY
b.book_id;
SELECT name FROM authors WHERE name LIKE 'Jr%' OR name LIKE 'Sr%' OR name LIKE 'III%' OR name LIKE 'Inc%'
LIMIT 10;
PRAGMA integrity_check; PRAGMA foreign_key_check;"
```

```
Exit code 1
Parse error near line 11 of load_books.sql: no such table: temp.staging_books
```

CLAUDE

`.import` doesn't accept the `temp.` prefix, so I'm switching that line to `--schema temp` and rebuilding.

Edit

```
C:\Users\johnh\workspace\language-of-computation\week2-assignment\load_books.sql
--- replace ---
.import --csv books.csv temp.staging_books
--- with ---
.import --csv --schema temp books.csv staging_books
```

Bash · Rebuild library.db and verify contents

```
cd "C:/Users/johnh/workspace/language-of-computation/week2-assignment" &&
S="/c/Users/johnh/AppData/Local/Microsoft/WinGet/Packages/SQLite.SQLite_Microsoft.Winget.Source_8wekyb3d8bb
we/sqlite3.exe" && rm -f library.db && "$S" -bail library.db ".read schema.sql" ".read load_books.sql" &&
"$S" -header -column library.db "
SELECT (SELECT count(*) FROM books) books, (SELECT count(*) FROM authors) authors, (SELECT count(*) FROM
book_authors) links, (SELECT count(*) FROM books WHERE isbn IS NULL) no_isbn;
SELECT b.book_id, b.isbn, b.title, b.publication_year yr, group_concat(a.name, ', ') authors FROM books b
JOIN book_authors ba USING(book_id) JOIN authors a USING(author_id) WHERE b.book_id <= 5 GROUP BY
b.book_id;
SELECT name FROM authors WHERE name LIKE 'Jr%' OR name LIKE 'Sr%' OR name LIKE 'III%' OR name LIKE 'Inc%'
LIMIT 10;
```

```
SELECT count(*) books_without_author FROM books WHERE book_id NOT IN (SELECT book_id FROM book_authors);
PRAGMA integrity_check; PRAGMA foreign_key_check;" && ls -l library.db
```

```
The file C:\Users\johnh\workspace\language-of-computation\week2-assignment\load_books.sql has been updated
successfully. (file state is current in your context — no need to Read it back)
```

```
books  authors  links  no_isbn
-----  -------  -----  -------
10000     5841  13209      700
book_id     isbn                              title                                yr             authors
-------  ----------  ----------------------------------------------------------  ----  --------------------------
      1  0439023483  The Hunger Games (The Hunger Games, #1)                     2008  Suzanne Collins
      2  0439554934  Harry Potter and the Sorcerer's Stone (Harry Potter, #1)     1997  J.K. Rowling, Mary GrandPré
      3  0316015849  Twilight (Twilight, #1)                                     2005  Stephenie Meyer
      4  0061120081  To Kill a Mockingbird                                       1960  Harper Lee
      5  0743273567  The Great Gatsby                                            1925  F. Scott Fitzgerald
books_without_author
--------------------
                   0
integrity_check
---------------
ok
-rw-r--r-- 1 johnh 197609 3145728 Oct 22 18:35 library.db
```

CLAUDE

The catalog loaded cleanly. Next I'll check that the lending rules actually block bad data, using a throwaway copy of the database.

Bash · Test constraints on a scratch copy of the database

```
cd "C:/Users/johnh/workspace/language-of-computation/week2-assignment" &&
S="/c/Users/johnh/AppData/Local/Microsoft/WinGet/Packages/SQLite.SQLite_Microsoft.Winget.Source_8wekyb3d8bb
we/sqlite3.exe" && T="C:/Users/johnh/AppData/Local/Temp/claude/C--Users-johnh-workspace/e0ec803d-aedf-42de-
955a-61f4ae0a2e79/scratchpad/t.db" && cp library.db "$T" && "$S" "$T" "PRAGMA foreign_keys=ON; INSERT INTO
members(first_name,last_name) VALUES('A','B'); INSERT INTO copies(book_id) VALUES(1); INSERT INTO
loans(copy_id,member_id) VALUES(1,1); SELECT 'loan ok', due_on FROM loans;" ; for q in "INSERT INTO
loans(copy_id,member_id) VALUES(1,1);" "INSERT INTO copies(book_id) VALUES(99999);" "INSERT INTO
copies(book_id,status) VALUES(1,'stolen');"; do "$S" "$T" "PRAGMA foreign_keys=ON; $q" 2>&1; done; rm "$T"
```

```
loan ok|2026-11-12
Error in 2nd command line argument: UNIQUE constraint failed: loans.copy_id
Error in 2nd command line argument: FOREIGN KEY constraint failed
Error in 2nd command line argument: CHECK constraint failed: status IN ('available', 'on_loan', 'lost', 'retired')
```

CLAUDE

I created library.db in your `week2-assignment` folder and filled its catalog with the 10,000 books from `books.csv`.

What's in it

Table	Holds	Rows now
`books`	One row per title: ISBN, title, year, language, rating	10,000
`authors`	One row per author	5,841
`book_authors`	Which authors wrote which book, since a book can have several	13,209
`members`	Your library's patrons	0
`copies`	The physical copies you own of each title, with condition and status	0

Table	Holds	Rows now
`loans`	Who borrowed which copy, the due date (21 days by default) and when it came back	0

The last three tables are empty because those records are yours to add as you sign up members and shelve books. The database itself passed SQLite's built-in checks.

What I fixed while loading

- **ISBNs:** I restored the leading zeros the CSV had dropped, so `61120081` is now `0061120081`. 700 books have no ISBN at all.
- **Years:** They're whole numbers now (`2008`, not `2008.0`).
- **Authors:** Lists like "J.K. Rowling, Mary GrandPré" are split into separate authors.
- **`isbn13`:** I left this column out, because the CSV had already rounded those numbers and they can't be recovered.

Rules the database enforces

- A copy can't be out on two loans at once.
- A copy can't refer to a book that isn't in the catalog.
- A copy's status can only be available, on loan, lost or retired.

I tested each of these on a throwaway copy, and each blocked bad data.

Using it: SQLite only checks the links between tables, such as a copy's book being in the catalog, when you run `PRAGMA foreign_keys = ON;` after opening the database. Run that at the start of every session.

Rebuilding it: The table design is in schema.sql and the import is in load_books.sql. To recreate the database from scratch, delete `library.db` and run:

```
sqlite3 library.db ".read schema.sql" ".read load_books.sql"
```

YOU · OCT 22, 2026, 6:36 PM

Commit library.db, schema.sql, load_books.sql and books.csv to the assignment branch and push to GitHub.

Bash · Commit four files and push assignment branch

```
cd "C:/Users/johnh/workspace/language-of-computation/week2-assignment" && git branch --show-current && git add library.db schema.sql load_books.sql books.csv && git commit -q -m "Add library database, schema, and goodbooks-10k import

Co-Authored-By: Claude Opus 5.5 <noreply@anthropic.com>" && git push 2>&1 | tail -2 && git status -sb && git log --oneline -3
```

```
assignment
warning: in the working copy of 'books.csv', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'load_books.sql', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'schema.sql', LF will be replaced by CRLF the next time Git touches it
To https://github.com/yuexinghu02/LoC-week2-assignment.git
   fa30544..a15dd9e  assignment -> assignment
## assignment...origin/assignment
a15dd9e Add library database, schema, and goodbooks-10k import
fa30544 Add README for Module 2 assignment
```

CLAUDE

I committed the four files to the `assignment` branch and pushed them to GitHub:
https://github.com/yuexinghu02/LoC-week2-assignment/tree/assignment

The commit is `a15dd9e` and contains `library.db`, `schema.sql`, `load_books.sql` and `books.csv`. Your local branch and GitHub match, and `main` still has only the README.

YOU · OCT 22, 2026, 6:36 PM

load at least 25 books from books.csv into my database, with their authors

Bash · Confirm books and authors are loaded

```
cd "C:/Users/johnh/workspace/language-of-computation/week2-assignment" &&
S="/c/Users/johnh/AppData/Local/Microsoft/WinGet/Packages/SQLite.SQLite_Microsoft.Winget.Source_8wekyb3d8bbwe/sqlite3.exe" && "$S" -header -column library.db "
SELECT (SELECT count(*) FROM books) books, (SELECT count(*) FROM authors) authors, (SELECT count(*) FROM books WHERE book_id NOT IN (SELECT book_id FROM book_authors)) books_missing_authors;
SELECT b.book_id, b.title, group_concat(a.name, ', ') authors FROM books b JOIN book_authors ba USING(book_id) JOIN authors a USING(author_id) WHERE b.book_id <= 25 GROUP BY b.book_id ORDER BY b.book_id;"
```

```
books  authors  books_missing_authors
-----  -------  --------------------
10000     5841                     0
book_id                           title                                                       authors
-------  ----------------------------------------------------------  ------------------------------------------
----------
      1  The Hunger Games (The Hunger Games, #1)                     Suzanne Collins
      2  Harry Potter and the Sorcerer's Stone (Harry Potter, #1)    J.K. Rowling, Mary GrandPré
      3  Twilight (Twilight, #1)                                     Stephenie Meyer
      4  To Kill a Mockingbird                                       Harper Lee
      5  The Great Gatsby                                            F. Scott Fitzgerald
      6  The Fault in Our Stars                                      John Green
      7  The Hobbit                                                  J.R.R. Tolkien
      8  The Catcher in the Rye                                      J.D. Salinger
      9  Angels & Demons  (Robert Langdon, #1)                       Dan Brown
     10  Pride and Prejudice                                         Jane Austen
     11  The Kite Runner                                             Khaled Hosseini
     12  Divergent (Divergent, #1)                                   Veronica Roth
     13  1984                                                        Celâl Üster, Erich Fromm, George Or
â€¦ [truncated, 1176 more characters]
```

CLAUDE

You already have well over 25 books. When I built `library.db`, I loaded all 10,000 books from `books.csv`, and every one is linked to at least one author. Here are the first 25:

#	Title	Authors
1	The Hunger Games	Suzanne Collins
2	Harry Potter and the Sorcerer's Stone	J.K. Rowling, Mary GrandPré
3	Twilight	Stephenie Meyer
4	To Kill a Mockingbird	Harper Lee
5	The Great Gatsby	F. Scott Fitzgerald
6	The Fault in Our Stars	John Green
7	The Hobbit	J.R.R. Tolkien
8	The Catcher in the Rye	J.D. Salinger
9	Angels & Demons	Dan Brown

#	Title	Authors
10	Pride and Prejudice	Jane Austen
11	The Kite Runner	Khaled Hosseini
12	Divergent	Veronica Roth
13	1984	Celâl Üster, Erich Fromm, George Orwell
14	Animal Farm	George Orwell
15	The Diary of a Young Girl	Anne Frank, B.M. Mooyaart-Doubleday, Eleanor Roosevelt
16	The Girl with the Dragon Tattoo	Reg Keeland, Stieg Larsson
17	Catching Fire	Suzanne Collins
18	Harry Potter and the Prisoner of Azkaban	J.K. Rowling, Mary GrandPré, Rufus Beck
19	The Fellowship of the Ring	J.R.R. Tolkien
20	Mockingjay	Suzanne Collins
21	Harry Potter and the Order of the Phoenix	J.K. Rowling, Mary GrandPré
22	The Lovely Bones	Alice Sebold
23	Harry Potter and the Chamber of Secrets	J.K. Rowling, Mary GrandPré
24	Harry Potter and the Goblet of Fire	J.K. Rowling, Mary GrandPré
25	Harry Potter and the Deathly Hallows	J.K. Rowling, Mary GrandPré

The dataset lists illustrators, translators and writers of introductions as authors too. For example, Mary GrandPré illustrated Harry Potter, and Erich Fromm wrote the afterword to *1984*. That's how the original data is, not a loading error.

I didn't change anything, since the database already does what you asked.

YOU · OCT 22, 2026, 6:38 PM

since my library doesnt have members yet, go ahead and make up at least 5 member and at least 10 borrowings. Include at least one book this is still out, one that is overdue, and one member who has never borrowed anything

Write
C:\Users\johnh\workspace\language-of-computation\week2-assignment\seed_data.sql

```
-- Made-up members, shelf copies and loans for library.db.
-- Run after schema.sql and load_books.sql: sqlite3 library.db ".read seed_data.sql"
-- Dates are fixed, so "overdue" depends on when you query: as of 2026-10-22,
-- loan 11 is overdue and loans 12 and 13 are still out but not yet due.

PRAGMA foreign_keys = ON;

BEGIN;

-- Sam Rivera (member 6) has never borrowed anything.
INSERT INTO members (member_id, first_name, last_name, email, phone, joined_on) VALUES
    (1, 'Maria', 'Lopez',     'maria.lopez@example.com',     '555-0101', '2026-07-20'),
    (2, 'James', 'Okafor',    'james.okafor@example.com',    '555-0102', '2026-07-22'),
    (3, 'Priya', 'Natarajan', 'priya.natarajan@example.com', '555-0103', '2026-08-01'),
```

```
    (4, 'Tom',   'Becker',    'tom.becker@example.com',      '555-0104', '2026-08-15'),
    (5, 'Aiko',  'Tanaka',    'aiko.tanaka@example.com',     '555-0105', '2026-08-30'),
    (6, 'Sam',   'Rivera',    'sam.rivera@example.com',      '555-0106', '2026-10-01');

-- One physical copy each of ten catalog titles.
INSERT INTO copies (copy_id, book_id, acquired_on, condition, status) VALUES
    (1,  1,  '2026-07-15', 'good', 'available'),  -- The Hunger Games
    (2,  2,  '2026-07-15', 'worn', 'available'),  -- Harry Potter and the Sorcerer's Stone
    (3,  4,  '2026-07-15', 'good', 'on_loan'),    -- To Kill a Mockingbird
    (4,  5,  '2026-07-15', 'good', 'available'),  -- The Great Gatsby
    (5,  7,  '2026-07-15', 'new',  'available'),  -- The Hobbit
    (6,  10, '2026-07-15', 'good', 'available'),  -- Pride and Prejudice
    (7,  11, '2026-07-15', 'good', 'available'),  -- The Kite Runner
    (8,  14, '2026-07-15', 'new',  'available'),  -- Animal Farm
    (9,  19, '2026-07-15', 'good', 'on_loan'),    -- The Fellowship of the Ring
    (10, 22, '2026-07-15', 'new',  'on_loan');    -- The Lovely Bones

INSERT INTO loans (loan_id, copy_id, member_id, loaned_on, due_on, returned_on) VALUES
    (1,  1,  1, '2026-08-03', '2026-08-24', '2026-08-20'),
    (2,  2,  2, '2026-08-05', '2026-08-26', '2026-08-25'),
    (3,  3,  3, '2026-08-12', '2026-09-02', '2026-09-05'),  -- returned 3 days late
    (4,  4,  1, '2026-08-25', '2026-09-15', '2026-09-10'),
    (5,  1,  4, '2026-09-01', '2026-09-22', '2026-09-18'),
    (6,  5,  5, '2026-09-03', '2026-09-24', '2026-09-20'),
    (7,  6,  2, '2026-09-10', '2026-10-01', '2026-09-28'),
    (8,  7,  3, '2026-09-15', '2026-10-
â€¦ [truncated, 364 more characters]
```

```
File created successfully at: C:\Users\johnh\workspace\language-of-computation\week2-assignment\seed_data.sql (file state is current in your context — no need to Read it back)
```

Bash · Load seed data and verify loans

```
cd "C:/Users/johnh/workspace/language-of-computation/week2-assignment" && S="/c/Users/johnh/AppData/Local/Microsoft/WinGet/Packages/SQLite.SQLite_Microsoft.Winget.Source_8wekyb3d8bbwe/sqlite3.exe" && "$S" library.db "SELECT count(*) FROM members; SELECT count(*) FROM loans;" && "$S" -bail library.db ".read seed_data.sql" && "$S" -header -column library.db "
PRAGMA foreign_keys=ON; PRAGMA foreign_key_check;
SELECT l.loan_id, m.first_name || ' ' || m.last_name member, b.title, l.loaned_on, l.due_on, l.returned_on,
  CASE WHEN l.returned_on IS NOT NULL THEN 'returned'
       WHEN l.due_on < date('now') THEN 'OVERDUE' ELSE 'out' END state
FROM loans l JOIN members m USING(member_id) JOIN copies c USING(copy_id) JOIN books b USING(book_id) ORDER BY l.loan_id;
SELECT m.first_name || ' ' || m.last_name never_borrowed FROM members m WHERE member_id NOT IN (SELECT member_id FROM loans);
SELECT c.copy_id, c.status, (SELECT count(*) FROM loans l WHERE l.copy_id=c.copy_id AND returned_on IS NULL) open_loans FROM copies c WHERE (c.status='on_loan') <> (open_loans>0);"
```

```
0
0
loan_id      member                              title                             loaned_on     due_on        returned_on   state
-------  ---------------  ----------------------------------------------------------  ----------  ----------  ----------  --------
      1  Maria Lopez      The Hunger Games (The Hunger Games, #1)                   2026-08-03  2026-08-24  2026-08-20  returned
      2  James Okafor     Harry Potter and the Sorcerer's Stone (Harry Potter, #1)  2026-08-05  2026-08-26  2026-08-25  returned
      3  Priya Natarajan  To Kill a Mockingbird                                     2026-08-12  2026-09-02  2026-
```

```
09-05   returned
      4  Maria Lopez      The Great Gatsby                                    2026-08-25  2026-09-15  2026-
09-10   returned
      5  Tom Becker       The Hunger Games (The Hunger Games, #1)             2026-09-01  2026-09-22  2026-
09-18   returned
      6  Aiko Tanaka      The Hobbit                                          2026-09-03  2026-09-24  2026-
09-20   returned
      7  James Okafor     Pride and Prejudice                                 2026-09-10  2026-10-01  2026-
09-28   returned
      8  Priya Natarajan  The Kite Runner                                     2026-09-15  2026-10-06  2026-
10-02   returned
      9  Tom Becker       Animal Farm                                         2026-09-20  2026-10-11  2026-
10-09   returned
     10  Aiko Tanaka      Harry Potter and the Sorcerer
â€¦ [truncated, 500 more characters]
```

CLAUDE

I added 6 made-up members and 13 borrowings to `library.db`, covering all three cases you asked for. The data is in seed_data.sql, so you can reload it whenever you rebuild the database.

Members

Member	Borrowings
Maria Lopez	3
James Okafor	3
Priya Natarajan	2
Tom Becker	3
Aiko Tanaka	2
Sam Rivera	0 (has never borrowed anything)

The members' email addresses use `example.com`, which is reserved for examples, and their phone numbers are fake 555 numbers.

Borrowings

- **Returned:** 10 loans, between August and October. Priya returned *To Kill a Mockingbird* 3 days late.
- **Overdue:** Tom Becker has *The Fellowship of the Ring*. It was due October 13 and hasn't come back.
- **Still out, not yet due:**
  - James Okafor has *To Kill a Mockingbird*, due October 31.
  - Maria Lopez has *The Lovely Bones*, due November 5.

Copies: Loans have to point at physical copies, so I also added one copy each of 10 popular titles. The three copies currently out are marked as on loan, and that matches the open loans.

Dates: The dates are fixed. A loan counts as overdue when its due date is earlier than the day you run the query, so after October 31 James's loan will show as overdue too.

YOU · OCT 22, 2026, 6:39 PM

Commit library.db and seed_data.sql to the assignment branch and push to GitHub.

Bash · Commit seed data and database, push

```
cd "C:/Users/johnh/workspace/language-of-computation/week2-assignment" && git branch --show-current && git add library.db seed_data.sql && git commit -q -m "Add sample members, copies and loans
```

```
Co-Authored-By: Claude Opus 5.5 <noreply@anthropic.com>" && git push 2>&1 | tail -1 && git status -sb &&
git log --oneline -1
```

```
assignment
warning: in the working copy of 'seed_data.sql', LF will be replaced by CRLF the next time Git touches it
   a15dd9e..036e8d3  assignment -> assignment
## assignment...origin/assignment
036e8d3 Add sample members, copies and loans
```

CLAUDE

I committed `library.db` and `seed_data.sql` to the `assignment` branch and pushed them to GitHub: https://github.com/yuexinghu02/LoC-week2-assignment/tree/assignment

The commit is `036e8d3`, and your local branch matches GitHub.

YOU · OCT 22, 2026, 6:40 PM

show me an aggregation of books by status

Bash · Aggregate copies by status

```
cd "C:/Users/johnh/workspace/language-of-computation/week2-assignment" &&
S="/c/Users/johnh/AppData/Local/Microsoft/WinGet/Packages/SQLite.SQLite_Microsoft.Winget.Source_8wekyb3d8bb
we/sqlite3.exe" && "$S" -header -column library.db "
SELECT c.status, count(*) AS copies, group_concat(b.title, '; ') AS titles
FROM copies c JOIN books b USING (book_id)
GROUP BY c.status ORDER BY copies DESC;
SELECT CASE WHEN l.loan_id IS NULL THEN 'available'
            WHEN l.due_on < date('now') THEN 'overdue' ELSE 'on loan, not due' END AS state,
       count(*) AS copies
FROM copies c LEFT JOIN loans l ON l.copy_id = c.copy_id AND l.returned_on IS NULL
GROUP BY state ORDER BY copies DESC;
SELECT count(*) AS catalog_titles_without_copies FROM books WHERE book_id NOT IN (SELECT book_id FROM
copies);"
```

```
status    copies                                                                                                titles
---------  ------  ------------------------------------------------------------------------------------------------------------
-------------------------------------------------------------------------------
available       7  The Hunger Games (The Hunger Games, #1); Harry Potter and the Sorcerer's Stone (Harry Potter,
#1); The Great Gatsby; The Hobbit; Pride and Prejudice; The Kite Runner; Animal Farm
on_loan         3  To Kill a Mockingbird; The Fellowship of the Ring (The Lord of the Rings, #1); The Lovely Bones
     state        copies
---------------  ------
available             7
on loan, not due      2
overdue               1
catalog_titles_without_copies
-----------------------------
                         9990
```

CLAUDE

Status is recorded on each physical copy, not on the catalog entry. Of your 10 shelf copies, 7 are available and 3 are on loan:

Status	Copies	Titles
available	7	The Hunger Games; Harry Potter and the Sorcerer's Stone; The Great Gatsby; The Hobbit; Pride and Prejudice; The Kite Runner; Animal Farm
on_loan	3	To Kill a Mockingbird; The Fellowship of the Ring; The Lovely Bones

```
SELECT c.status, count(*) AS copies, group_concat(b.title, '; ') AS titles
FROM copies c JOIN books b USING (book_id)
GROUP BY c.status
ORDER BY copies DESC;
```

The `status` column doesn't record whether a loan is overdue. Working that out from today's date (October 22) and the open loans splits the 3 out on loan further:

State	Copies
available	7
on loan, not due	2
overdue	1 (The Fellowship of the Ring)

```
SELECT CASE WHEN l.loan_id IS NULL THEN 'available'
            WHEN l.due_on < date('now') THEN 'overdue'
            ELSE 'on loan, not due' END AS state,
       count(*) AS copies
FROM copies c
LEFT JOIN loans l ON l.copy_id = c.copy_id AND l.returned_on IS NULL
GROUP BY state
ORDER BY copies DESC;
```

The other 9,990 titles in the catalog have no copies on your shelves, so they don't appear in either breakdown.

YOU · OCT 22, 2026, 6:41 PM

Read the schema of library.db as it is now, and draw an ERD of it as a Mermaid diagram, and create a PNG. Put it at the top of a new file named summary.md

Bash · Dump schema, foreign keys, and check rendering tools

```
cd "C:/Users/johnh/workspace/language-of-computation/week2-assignment" &&
S="/c/Users/johnh/AppData/Local/Microsoft/WinGet/Packages/SQLite.SQLite_Microsoft.Winget.Source_8wekyb3d8bbwe/sqlite3.exe" && "$S" library.db ".schema" && echo --- && "$S" library.db "SELECT m.name, p.* FROM sqlite_master m, pragma_foreign_key_list(m.name) p WHERE m.type='table';" && echo --- && which node npx mmdc 2>&1 | grep -v 'no '; node --version 2>&1; ls "/c/Program Files/Google/Chrome/Application/chrome.exe" "/c/Program Files (x86)/Microsoft/Edge/Application/msedge.exe" 2>&1
```

```
CREATE TABLE books (
    book_id           INTEGER PRIMARY KEY,
    goodreads_book_id INTEGER UNIQUE,
    isbn              TEXT,
    title             TEXT NOT NULL,
    original_title    TEXT,
    publication_year  INTEGER,
    language_code     TEXT,
    average_rating    REAL CHECK (average_rating BETWEEN 0 AND 5),
    ratings_count     INTEGER CHECK (ratings_count >= 0),
    image_url         TEXT
```

```
);
CREATE TABLE authors (
    author_id INTEGER PRIMARY KEY,
    name      TEXT NOT NULL UNIQUE
);
CREATE TABLE book_authors (
    book_id      INTEGER NOT NULL REFERENCES books (book_id) ON DELETE CASCADE,
    author_id    INTEGER NOT NULL REFERENCES authors (author_id),
    author_order INTEGER NOT NULL DEFAULT 1,
    PRIMARY KEY (book_id, author_id)
);
CREATE TABLE members (
    member_id  INTEGER PRIMARY KEY,
    first_name TEXT NOT NULL,
    last_name  TEXT NOT NULL,
    email      TEXT UNIQUE,
    phone      TEXT,
    joined_on  TEXT NOT NULL DEFAULT (date('now'))
);
CREATE TABLE copies (
    copy_id     INTEGER PRIMARY KEY,
    book_id     INTEGER NOT NULL REFERENCES books (book_id),
    acquired_on TEXT NOT NULL DEFAULT (date('now')),
    condition   TEXT NOT NULL DEFAULT 'good'
                CHECK (condition IN ('new', 'good', 'worn', 'damaged')),
    status      TEXT NOT NULL DEFAULT 'available'
                CHECK (status IN ('available', 'on_loan', 'lost', 'retired'))
);
CREATE TABLE loans (
    loan_id     INTEGER PRIMARY
â€¦ [truncated, 1177 more characters]
```

CLAUDE

No Node or Mermaid CLI is installed, so I'll render the PNG with headless Chrome. Chrome will load the Mermaid library from the jsDelivr CDN, and the diagram itself never leaves this computer.

Write
C:\Users\johnh\workspace\language-of-computation\week2-assignment\erd.mmd

```
erDiagram
    books ||--o{ book_authors : "written by"
    authors ||--o{ book_authors : "writes"
    books ||--o{ copies : "shelved as"
    copies ||--o{ loans : "lent in"
    members ||--o{ loans : "borrows"

    books {
        INTEGER book_id PK
        INTEGER goodreads_book_id UK
        TEXT isbn
        TEXT title "NOT NULL"
        TEXT original_title
        INTEGER publication_year
        TEXT language_code
        REAL average_rating "0 to 5"
        INTEGER ratings_count ">= 0"
        TEXT image_url
```

```
    }
    authors {
        INTEGER author_id PK
        TEXT name UK "NOT NULL"
    }
    book_authors {
        INTEGER book_id PK, FK "ON DELETE CASCADE"
        INTEGER author_id PK, FK
        INTEGER author_order "default 1"
    }
    members {
        INTEGER member_id PK
        TEXT first_name "NOT NULL"
        TEXT last_name "NOT NULL"
        TEXT email UK
        TEXT phone
        TEXT joined_on "default today"
    }
    copies {
        INTEGER copy_id PK
        INTEGER book_id FK "NOT NULL"
        TEXT acquired_on "default today"
        TEXT condition "new, good, worn, damaged"
        TEXT status "available, on_loan, lost, retired"
    }
    loans {
        INTEGER loan_id PK
        INTEGER copy_id FK "one open loan per copy"
        INTEGER member_id FK "NOT NULL"
        TEXT loaned_on "default today"
        TEXT due_on "default today + 21 days"
        TEXT returned_on "NULL while out"
    }
```

```
File created successfully at: C:\Users\johnh\workspace\language-of-computation\week2-assignment\erd.mmd (file state is current in your context — no need to Read it back)
```

Bash · Build render page and measure diagram size in headless Chrome

```
cd "C:/Users/johnh/workspace/language-of-computation/week2-assignment" &&
P="C:/Users/johnh/AppData/Local/Temp/claude/C--Users-johnh-workspace/e0ec803d-aedf-42de-955a-
61f4ae0a2e79/scratchpad" && { cat <<'EOF'
<!doctype html><html><head><meta charset="utf-8">
<style>html,body{margin:0;background:#fff}#d{display:inline-block;padding:24px}</style>
<script src="https://cdn.jsdelivr.net/npm/mermaid@10.9.1/dist/mermaid.min.js"></script></head><body>
<div id="d"><pre class="mermaid">
EOF
sed -e 's/&/\&amp;/g' -e 's/</\&lt;/g' -e 's/>/\&gt;/g' erd.mmd
cat <<'EOF'
</pre></div>
<script>
mermaid.initialize({startOnLoad:false, theme:'default', er:{useMaxWidth:false}});
mermaid.run().then(()=>{const
r=document.getElementById('d').getBoundingClientRect();document.body.setAttribute('data-
size',Math.ceil(r.width)+'x'+Math.ceil(r.height));});
</script></body></html>
EOF
} > "$P/erd.html" && "/c/Program Files/Google/Chrome/Application/chrome.exe" --headless=new --disable-gpu -
```

```
-virtual-time-budget=15000 --window-size=3000,3000 --dump-dom "file:///$P/erd.html" 2>/dev/null | grep -o
'data-size="[0-9x]*"'
```

```
data-size="1066x873"
```

Bash · Screenshot the rendered ERD to erd.png at 2x

```
cd "C:/Users/johnh/workspace/language-of-computation/week2-assignment" &&
P="C:/Users/johnh/AppData/Local/Temp/claude/C--Users-johnh-workspace/e0ec803d-aedf-42de-955a-
61f4ae0a2e79/scratchpad" && "/c/Program Files/Google/Chrome/Application/chrome.exe" --headless=new --
disable-gpu --hide-scrollbars --force-device-scale-factor=2 --virtual-time-budget=15000 --window-
size=1066,873 --screenshot="C:/Users/johnh/workspace/language-of-computation/week2-assignment/erd.png"
"file:///$P/erd.html" 2>/dev/null; ls -l erd.png
```

```
-rw-r--r-- 1 johnh 197609 147661 Oct 22 18:42 erd.png
```

Read

```
{"file_path":"C:\\Users\\johnh\\workspace\\language-of-computation\\week2-assignment\\erd.png"}
```

```
[image]
```

CLAUDE

The PNG renders cleanly. Now I'll write `summary.md` with the diagram at the top.

Write

C:\Users\johnh\workspace\language-of-computation\week2-assignment\summary.md

````
# library.db Summary

## Entity-Relationship Diagram

```mermaid
erDiagram
    books ||--o{ book_authors : "written by"
    authors ||--o{ book_authors : "writes"
    books ||--o{ copies : "shelved as"
    copies ||--o{ loans : "lent in"
    members ||--o{ loans : "borrows"

    books {
        INTEGER book_id PK
        INTEGER goodreads_book_id UK
        TEXT isbn
        TEXT title "NOT NULL"
        TEXT original_title
        INTEGER publication_year
        TEXT language_code
        REAL average_rating "0 to 5"
        INTEGER ratings_count ">= 0"
        TEXT image_url
    }
    authors {
        INTEGER author_id PK
        TEXT name UK "NOT NULL"
    }
    book_authors {
        INTEGER book_id PK, FK "ON DELETE CASCADE"
        INTEGER author_id PK, FK
````
````

```
        INTEGER author_order "default 1"
    }
    members {
        INTEGER member_id PK
        TEXT first_name "NOT NULL"
        TEXT last_name "NOT NULL"
        TEXT email UK
        TEXT phone
        TEXT joined_on "default today"
    }
    copies {
        INTEGER copy_id PK
        INTEGER book_id FK "NOT NULL"
        TEXT acquired_on "default today"
        TEXT condition "new, good, worn, damaged"
        TEXT status "available, on_loan, lost, retired"
    }
    loans {
        INTEGER loan_id PK
        INTEGER copy_id FK "one open loan per copy"
        INTEGER member_id FK "NOT NULL"
        TEXT loaned_on "default today"
        TEXT due_on "default today + 21 days"
        TEXT returned_on "NULL while out"
    }
```

PNG version ([erd.png](erd.png)):



Tables

Table	Purpose	Rows
`books`	Catalog of titles, loaded from goodbooks-10k `books.csv`	10,000
`authors`	One row per author	5,841
`book_authors`	Links books to their authors (many-to-many)	13,209
`members`	Library patrons (made-up sample data)	6
`copies`	Physical copies on the shelves	10
`loans`	Borrowing history and open loans	13

Relationships

- A **book** has one or more **authors**, and an author can write many books, linked through `book_authors`.
- A **book** can have many physical **copies**; each copy belongs to exactly one book.
- A **member** can have many **loans**, and a **copy** can be lent many times, but only one open loan per copy at a time (`idx_loans_one_open_per_copy`).

Files

- `schema.sql` creates the tables and indexes.
â€¦ [truncated, 202 more characters]

File created successfully at: C:\Users\johnh\workspace\language-of-computation\week2-assignment\summary.md (file state is current in your context — no need to Read it back)

CLAUDE

I read the schema straight from `library.db`, drew it as a Mermaid ERD, rendered it to erd.png, and put both at the top of a new summary.md.

What the diagram shows: all 6 tables with their columns, types, keys and constraints. It also shows the 5 relationships between them:

- authors write books, linked through `book_authors`
- each book can have many copies on the shelves
- each copy can be lent many times
- each member can borrow many times

`summary.md` has the Mermaid diagram first, then the PNG. GitHub draws the Mermaid version itself, so the PNG is mainly for places that can't. Below the diagram is a short section listing each table's purpose and row count, the relationships, and the project files.

The PNG: I rendered it with headless Chrome, which loaded the Mermaid library from the jsDelivr CDN; the diagram itself never left this computer. It's 2132×1746 pixels, so it stays sharp when you zoom in. I checked the image and everything is readable.

`erd.mmd` holds the Mermaid source on its own, so you can edit the diagram and re-render it later.

YOU · OCT 22, 2026, 6:42 PM

Commit summary.md, erd.png and erd.mmd to the assignment branch and push to GitHub.

Bash · Commit ERD and summary, push

```
cd "C:/Users/johnh/workspace/language-of-computation/week2-assignment" && git branch --show-current && git add summary.md erd.png erd.mmd && git commit -q -m "Add ERD diagram and database summary

Co-Authored-By: Claude Opus 5.5 <noreply@anthropic.com>" && git push 2>&1 | tail -1 && git status -sb && git log --oneline -1
```

```
assignment
warning: in the working copy of 'erd.mmd', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'summary.md', LF will be replaced by CRLF the next time Git touches it
   036e8d3..f9b7b0c  assignment -> assignment
## assignment...origin/assignment
f9b7b0c Add ERD diagram and database summary
```

CLAUDE

I committed `summary.md`, `erd.png` and `erd.mmd` to the `assignment` branch and pushed them to GitHub. The commit is `f9b7b0c`, and your local branch matches GitHub.

You can see the summary, with the diagram drawn at the top, here: https://github.com/yuexinghu02/LoC-week2-assignment/blob/assignment/summary.md

YOU · OCT 22, 2026, 6:45 PM

write a summary using the summary.md folder. Write the summary below the PNG that is one page long in your own words. Include:

- What each table represents, and why it exists.
- Where your one-to-many and many-to-many relationships are, and how the database keeps track of them.
- One design decision you made and why (for example, how you handled multiple copies, multiple authors, or lost books).
- One thing Claude got wrong or that you changed.

Read

```
{"file_path":"C:\\Users\\johnh\\workspace\\language-of-computation\\week2-assignment\\summary.md","offset":60}
```

```
60
61      ![ERD of library.db](erd.png)
62
63      ## Tables
64
65      | Table | Purpose | Rows |
66      |---|---|---|
67      | `books` | Catalog of titles, loaded from goodbooks-10k `books.csv` | 10,000 |
68      | `authors` | One row per author | 5,841 |
69      | `book_authors` | Links books to their authors (many-to-many) | 13,209 |
70      | `members` | Library patrons (made-up sample data) | 6 |
71      | `copies` | Physical copies on the shelves | 10 |
72      | `loans` | Borrowing history and open loans | 13 |
73
74      ## Relationships
75
76      - A **book** has one or more **authors**, and an author can write many books, linked through `book_authors`.
77      - A **book** can have many physical **copies**; each copy belongs to exactly one book.
78      - A **member** can have many **loans**, and a **copy** can be lent many times, but only one open loan per copy at a time (`idx_loans_one_open_per_copy`).
79
80      ## Files
81
82      - `schema.sql` creates the tables and indexes.
83      - `load_books.sql` imports `books.csv` into `books`, `authors` and `book_authors`.
84      - `seed_data.sql` adds the sample members, copies and loans.
85      - `erd.mmd` is the Mermaid source for the diagram above.
86
```

Bash · Replace content below the PNG with the written summary

```
cd "C:/Users/johnh/workspace/language-of-computation/week2-assignment" && head -n 62 summary.md > summary.tmp && cat >> summary.tmp <<'EOF'
## Summary

### What each table represents

`library.db` models a small lending library with six tables. **`books`** is the catalog: one row per title, with its ISBN, publication year, language and Goodreads rating. I loaded all 10,000 titles from the goodbooks-10k dataset, so it describes every book the library *could* own, not what is on the shelf. **`authors`** stores each author's name exactly once, so "J.K. Rowling" is not repeated across every Harry
```

```
Potter row. **`book_authors`** connects the two and exists only because a book can have more than one author. **`members`** holds the people who hold a library card. **`copies`** represents the physical books on the shelves, each with its own condition and status. **`loans`** records every borrowing: which copy, which member, when it went out, when it is due, and when it came back.

### One-to-many and many-to-many relationships

There are three one-to-many relationships. One book can have many copies (`copies.book_id`), one copy can be lent many times over its life (`loans.copy_id`), and one member can borrow many times (`loans.member_id`). In each case the "many" side stores a foreign key pointing back to the "one" side's primary key, and SQLite refuses a copy or loan that points at a row that doesn't exist (as long as `PRAGMA foreign_keys = ON` is set).

The many-to-many relationship is between books and authors: *1984* has several contributors listed, and George Orwell wrote both *1984* and *Animal Farm*. A single column can't hold that, so `book_authors` stores one row per (book, author) pair. Its primary key is the pair itself, which stops the same author being linked to the same book twice, and `author_order` remembers who is listed first.

### A design decision: separating titles from copies

The most important choice was splitting `books` from `copies`. If the status lived on the `books` table, the library could never own two copies of *The Hunger Games* and lend one while the other stays on the shelf. With a separate `copies` table, each physical book has its own `status` (available, on loan, lost or retired) and condition. This also handles lost books without deleting anything: a lost copy is marked `lost`, so its loan history stays intact. Finally, a partial unique index (`idx_loans_one_open_per_copy`) makes it impossible to record the same copy as checked
â€¦ [truncated, 859 more characters]
```

```
503
```

CLAUDE

I added a summary of about 500 words, roughly one page, under the PNG in summary.md. It has four sections, one for each point you listed:

1. **What each table represents:** what each of the 6 tables holds, and why it's a separate table.
2. **Relationships:**
   - The three one-to-many links, and how a foreign key on the "many" side keeps track of each one.
   - The books-to-authors many-to-many link, handled by `book_authors`.
3. **Design decision:** keeping titles (`books`) separate from physical copies (`copies`). That lets you own several copies of a title and mark a lost book without losing its loan history. A special index also stops one copy being lent to two people at once.
4. **What Claude got wrong:** two real problems from this project.
   - My first import script failed because SQLite's `.import` doesn't accept the `temp.` prefix, so I had to fix it and rebuild the database.
   - The dataset lists illustrators and translators as authors, for example Mary GrandPré on Harry Potter. The summary suggests adding a `role` column to tell them apart.

Two things to check before you submit:

- **Removed sections:** The new summary replaces the earlier Tables, Relationships and Files sections, because they repeated the same information. Their contents (the row counts, relationships and file list) are no longer in `summary.md`.
- **Your voice:** The assignment asks for this in your own words, and I wrote it in the first person as if you were the author. Read it through and change anything that doesn't match how you'd put it. The fourth

point especially should reflect what you noticed and changed yourself.

YOU · OCT 22, 2026, 6:45 PM

Commit summary.md to the assignment branch and push to GitHub.

Bash · Commit summary.md and push

```
cd "C:/Users/johnh/workspace/language-of-computation/week2-assignment" && git branch --show-current && git add summary.md && git commit -q -m "Add written one-page summary to summary.md

Co-Authored-By: Claude Opus 5.5 <noreply@anthropic.com>" && git push 2>&1 | tail -1 && git status -sb && git log --oneline -1
```

```
assignment
warning: in the working copy of 'summary.md', LF will be replaced by CRLF the next time Git touches it
   f9b7b0c..7ea312c  assignment -> assignment
## assignment...origin/assignment
7ea312c Add written one-page summary to summary.md
```

CLAUDE

I committed `summary.md` to the `assignment` branch and pushed it to GitHub. The commit is `7ea312c`, and your local branch matches GitHub.

You can read it on GitHub here: https://github.com/yuexinghu02/LoC-week2-assignment/blob/assignment/summary.md

YOU · OCT 22, 2026, 7:02 PM

practically, i want to be able to take that list of 10000 books and be able to sort the books in whatever specific way such as by author, date, condition, or even a combination of both or multiple

PowerShell · Check script execution policy and sqlite3 on PATH

```
Get-ExecutionPolicy -List | Format-Table -AutoSize | Out-String; (Get-Command sqlite3 -ErrorAction SilentlyContinue).Source
```

```
Scope ExecutionPolicy
        ----- ---------------
MachinePolicy       Undefined
   UserPolicy       Undefined
      Process          Bypass
  CurrentUser       Undefined
 LocalMachine       Undefined
```

Bash · Add book_catalog view to schema.sql and library.db

```
cd "C:/Users/johnh/workspace/language-of-computation/week2-assignment" && cat > view.tmp <<'EOF'

-- One row per physical copy, plus one row for each title with no copies (copy columns NULL),
-- with author and copy details flattened so the catalog can be sorted on any mix of columns.
CREATE VIEW book_catalog AS
SELECT b.book_id,
       b.title,
       pa.name AS primary_author,
       -- Last word of the primary author's name, for shelf-style sorting ("Tolkien" for "J.R.R. Tolkien").
       substr(pa.name, length(rtrim(pa.name, replace(pa.name, ' ', ''))) + 1) AS author_last_name,
       (SELECT group_concat(name, ', ')
```

```
        FROM (SELECT a.name FROM book_authors x JOIN authors a USING (author_id)
              WHERE x.book_id = b.book_id ORDER BY x.author_order)) AS all_authors,
       b.publication_year,
       b.language_code,
       b.average_rating,
       b.ratings_count,
       c.copy_id,
       c.condition,
       -- Best to worst, so sorting by condition isn't alphabetical.
       CASE c.condition WHEN 'new' THEN 1 WHEN 'good' THEN 2 WHEN 'worn' THEN 3 WHEN 'damaged' THEN 4 END
           AS condition_rank,
       c.status,
       c.acquired_on
FROM books b
LEFT JOIN book_authors ba ON ba.book_id = b.book_id AND ba.author_order = 1
LEFT JOIN authors pa ON pa.author_id = ba.author_id
LEFT JOIN copies c ON c.book_id = b.book_id;
EOF
cat view.tmp >> schema.sql &&
S="/c/Users/johnh/AppData/Local/Microsoft/WinGet/Packages/SQLite.SQLite_Microsoft.Winget.Source_8wekyb3d8bb
we/sqlite3.exe" && "$S" -bail library.db ".read view.tmp" && rm view.tmp && "$S" -header -column library.db
"SELECT count(*) rows_, count(copy_id) copy_rows, count(primary_author) with_author FROM book_catalog;
SELECT book_id, title, primary_author, author_last_name, all_authors FROM book_catalog WHERE book_id IN
(7,13,15);"
```

```
rows_  copy_rows  with_author
-----  ---------  -----------
10000         10        10000
book_id            title            primary_author  author_last_name                      all_authors
-------  -------------------------  --------------  ----------------  ------------------------------------------
---------
      7  The Hobbit                 J.R.R. Tolkien  Tolkien           J.R.R. Tolkien
     13  1984                       George Orwell   Orwell            George Orwell, Erich Fromm, Celâl Üster
     15  The Diary of a Young Girl  Anne Frank      Frank             Anne Frank, Eleanor Roosevelt, B.M. Mooyaart-
Doubleday
```

CLAUDE

Next is the sort command. PowerShell blocks `.ps1` scripts on this computer by default, so I'll add a `.cmd` wrapper that runs the script without changing any of your settings.

Write
C:\Users\johnh\workspace\language-of-computation\week2-assignment\sort-books.ps1

```
<#
.SYNOPSIS
    Sorts the library.db catalog by any combination of columns.

.EXAMPLE
    .\sort-books.cmd author year
    .\sort-books.cmd rating:desc popularity:desc -Limit 10
    .\sort-books.cmd condition title -OnShelf
    .\sort-books.cmd language author year:desc -Limit 0 -Csv sorted.csv

    Sort keys apply in the order given; add :desc to reverse one. Blank values
    (no year, no copy on the shelf) always sort last.
#>
```

```
param(
    [Parameter(Position = 0, ValueFromRemainingArguments = $true)]
    [string[]] $By = @('title'),
    # How many rows to show; 0 shows all of them.
    [int] $Limit = 25,
    # Only books the library has a physical copy of.
    [switch] $OnShelf,
    # Write the sorted rows to this CSV file instead of the screen.
    [string] $Csv
)

$ErrorActionPreference = 'Stop'

# Sort key => SQL expression(s) on the book_catalog view.
$sortKeys = [ordered]@{
    title      = 'title COLLATE NOCASE'
    author     = 'author_last_name COLLATE NOCASE, primary_author COLLATE NOCASE'
    authorfull = 'primary_author COLLATE NOCASE'
    year       = 'publication_year'
    rating     = 'average_rating'
    popularity = 'ratings_count'
    language   = 'language_code'
    condition  = 'condition_rank'
    status     = 'status'
    acquired   = 'acquired_on'
    id         = 'book_id'
}

# From cmd.exe, "author,year" arrives as one string, so split on commas as well as spaces.
$keys = $By | ForEach-Object { $_ -split ',' } | Where-Object { $_ -ne '' }

$orderBy = foreach ($key in $keys) {
    $name, $direction = $key.ToLower() -split ':', 2
    if (-not $sortKeys.Contains($name)) {
        throw "Unknown sort key '$name'. Valid keys: $($sortKeys.Keys -join ', ')"
    }
    if (-not $direction) { $direction = 'asc' }
    if ($direction -notin 'asc', 'desc') {
        throw "Unknown direction '$direction' in '$key'. Use :asc or :desc."
    }
    foreach ($expr in $sortKeys[$name] -split ', ') {
        "$expr $($direction.ToUpper()) NULLS LAST"
    }
}
# Tiebreakers so equal rows always come out in the same order.
$orderBy = @($orderBy) + 'book_id', 'copy_id'

$sql = @"
SELECT book_id, title, all_authors AS authors, publication_year AS year,
       average_rating AS rating, ratings_count AS ratings, language_code AS lang,
       condition, status
FROM book_catalog
$(if ($OnShelf) { 'WHERE copy_id IS NOT NULL' })
ORDER BY $($orderBy -join ', ')
$(if ($Limit -gt 0) {
â€¦ [truncated, 606 more characters]
```

Write
C:\Users\johnh\workspace\language-of-computation\week2-assignment\sort-books.cmd

```
@echo off
rem Runs sort-books.ps1 without changing the computer's PowerShell script policy.
powershell -NoProfile -ExecutionPolicy Bypass -File "%~dp0sort-books.ps1" %*
```

```
File created successfully at: C:\Users\johnh\workspace\language-of-computation\week2-assignment\sort-books.ps1
(file state is current in your context — no need to Read it back)
```

```
File created successfully at: C:\Users\johnh\workspace\language-of-computation\week2-assignment\sort-books.cmd
(file state is current in your context — no need to Read it back)
```

PowerShell · Test several sort combinations

```
Set-Location "C:\Users\johnh\workspace\language-of-computation\week2-assignment"; "== author year"; .\sort-books.cmd author year -Limit 6; "== rating:desc popularity:desc"; .\sort-books.cmd rating:desc popularity:desc -Limit 5; "== condition title -OnShelf"; .\sort-books.cmd condition title -OnShelf; "== status,condition (comma form)"; .\sort-books.cmd "status,condition:desc" -OnShelf -Limit 4; "== bad key"; .\sort-books.cmd pages
```

```
Exit code 1
== author year
book_id                   title                                    authors                  year  rating  ratings
lang  condition  status
-------  ---------------------------------------  ---------------------------------------  ----  ------  -------  -
---  ---------  ------
   5816  Why Mosquitoes Buzz in People's Ears     Verna Aardema, Leo Dillon, Diane Dillon  1975    4.01    20639
   2609  Midnight Riot (Peter Grant, #1)          Ben Aaronovitch                          2011    3.92    21519
eng
   4498  Moon Over Soho (Peter Grant, #2)         Ben Aaronovitch                          2011     4.1    16566
eng
   4922  Whispers Under Ground (Peter Grant, #3)  Ben Aaronovitch                          2012    4.16    15946
eng
   5956  Broken Homes (Peter Grant, #4)           Ben Aaronovitch                          2013    4.16    10958
eng
   6926  Foxglove Summer (Peter Grant, #5)        Ben Aaronovitch                          2014    4.18    12167
eng
== rating:desc popularity:desc
book_id                   title                                     authors                   year  rating  ratings
lang   condition  status
-------  ---------------------------------------  ---------------------------------------  ----  ------  -------
-----  ---------  ------
   3628  The Complete Calvin and Hobbes           Bill Watterson                           2005    4.82    28900
eng

    862  Words of Radiance (The Stormlight
â€¦ [truncated, 3756 more characters]
```

CLAUDE

Every sort combination works. Two small fixes remain: the bad-key error should be a clean message instead of a PowerShell stack trace, and I still need to test the CSV export.

Bash · Replace throws with clean error messages

```
cd "C:/Users/johnh/workspace/language-of-computation/week2-assignment" && sed -i -e "s/^        throw \"Unknown sort key '\$name'. Valid keys: \$(\$sortKeys.Keys -join ', ')\"/        Write-Host \"Unknown sort key '\$name'. Valid keys: \$(\$sortKeys.Keys -join ', ')\" -ForegroundColor Red\n        exit 1/" -e "s/^throw \"Unknown direction '\$direction' in '\$key'. Use :asc or :desc.\"/        Write-Host \"Unknown
```

```
direction '\$direction' in '\$key'. Use :asc or :desc.\" -ForegroundColor Red\n        exit 1/" sort-books.ps1 && grep -n -A1 'Write-Host "Unknown' sort-books.ps1
```

```
48:        Write-Host "Unknown sort key '$name'. Valid keys: $($sortKeys.Keys -join ', ')" -ForegroundColor Red
49-        exit 1
--
53:        Write-Host "Unknown direction '$direction' in '$key'. Use :asc or :desc." -ForegroundColor Red
54-        exit 1
```

PowerShell · Test bad-key message and full CSV export

```
Set-Location "C:\Users\johnh\workspace\language-of-computation\week2-assignment"; .\sort-books.cmd pages; "exit=$LASTEXITCODE"; $p = "C:\Users\johnh\AppData\Local\Temp\claude\C--Users-johnh-workspace\e0ec803d-aedf-42de-955a-61f4ae0a2e79\scratchpad\sorted.csv"; .\sort-books.cmd language author year:desc -Limit 0 -Csv $p; (Get-Content $p -Encoding UTF8 | Measure-Object -Line).Lines; Get-Content $p -Encoding UTF8 -TotalCount 4; Remove-Item $p
```

```
Exit code 1
Unknown sort key 'pages'. Valid keys: title, author, authorfull, year, rating, popularity, language, condition,
status, acquired, id
exit=1
C:\Users\johnh\workspace\language-of-computation\week2-assignment\sort-books.ps1 : Exception calling "GetFullPath"
with "1" argument(s): "The given path's format is not supported."
    + CategoryInfo          : NotSpecified: (:) [sort-books.ps1], MethodInvocationException
    + FullyQualifiedErrorId : NotSupportedException,sort-books.ps1

Get-Content : Cannot find path 'C:\Users\johnh\AppData\Local\Temp\claude\C--Users-johnh-workspace\e0ec803d-aedf-
42de-95
5a-61f4ae0a2e79\scratchpad\sorted.csv' because it does not exist.
At line:1 char:625
+ ... or year:desc -Limit 0 -Csv $p; (Get-Content $p -Encoding UTF8 | Measu ...
+                                    ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : ObjectNotFound: (C:\Users\johnh\...hpad\sorted.csv:String) [Get-Content],
ItemNotFoundEx
   ception
    + FullyQualifiedErrorId : PathNotFound,Microsoft.PowerShell.Commands.GetContentCommand

0
Get-Content : Cannot find path 'C:\Users\johnh\AppData\Local\Temp\claude\C--Users-johnh-workspace\e0ec803d-aedf-
42de-95
5a-61f4ae0a2e79\scratchpad\sorted.csv' because it does not exist.
At line:1 char:686
+ ... ect -Line).Lines; Get-Content $p -Encoding UTF8 -TotalCount 4; Remove ...
+                       ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : ObjectNotFound: (C:\Us
â€¦ [truncated, 761 more characters]
```

CLAUDE

The CSV export broke when given a full path, because `Join-Path` glued it onto the current folder. I'm switching to `Path.Combine`, which handles both relative and full paths.

Edit

```
C:\Users\johnh\workspace\language-of-computation\week2-assignment\sort-books.ps1
--- replace ---
    $out = [IO.Path]::GetFullPath((Join-Path (Get-Location) $Csv)) -replace '\\', '/'
--- with ---
    # Combine keeps $Csv as-is when it is already a full path.
    $out = [IO.Path]::GetFullPath([IO.Path]::Combine((Get-Location).Path, $Csv)) -replace '\\', '/'
```

PowerShell · Retest CSV export with full and relative paths

```
Set-Location "C:\Users\johnh\workspace\language-of-computation\week2-assignment"; $p =
"C:\Users\johnh\AppData\Local\Temp\claude\C--Users-johnh-workspace\e0ec803d-aedf-42de-955a-
61f4ae0a2e79\scratchpad\sorted.csv"; .\sort-books.cmd language author year:desc -Limit 0 -Csv $p; (Get-
Content $p -Encoding UTF8 | Measure-Object -Line).Lines; Get-Content $p -Encoding UTF8 -TotalCount 4;
Remove-Item $p; Push-Location $env:TEMP; & "C:\Users\johnh\workspace\language-of-computation\week2-
assignment\sort-books.cmd" year -Limit 2 -Csv rel-test.csv; Test-Path rel-test.csv; Remove-Item rel-
test.csv; Pop-Location
```

```
The file C:\Users\johnh\workspace\language-of-computation\week2-assignment\sort-books.ps1 has been updated
successfully. (file state is current in your context — no need to Read it back)
```

```
Wrote sorted rows to C:/Users/johnh/AppData/Local/Temp/claude/C--Users-johnh-workspace/e0ec803d-aedf-42de-955a-
61f4ae0a2e79/scratchpad/sorted.csv
10001
book_id,title,authors,year,rating,ratings,lang,condition,status
8634,"حوجن [Ḥawjan]","Ibraheem Abbas, ابراهيم عباس, Yasser Bahjatt",2013,3.75,10203,ara,,
6089,"الرحيق المختوم","Safiy al-Rahman al-Mubarakfuri",1979,4.57,10558,ara,,
2082,"The Qur'an / القرآن الكريم",Anonymous,609,4.34,31765,ara,,
Wrote sorted rows to C:/Users/johnh/AppData/Local/Temp/rel-test.csv
True
```

CLAUDE

You can now sort all 10,000 books by any column or any combination of columns. You list the sort keys in order, and the first one wins. Run it from your `week2-assignment` folder:

```
.\sort-books.cmd author year
```

That sorts by author last name, then by year within each author.

```
.\sort-books.cmd rating:desc popularity:desc -Limit 10
```

That shows the top 10 by rating, and when two books tie on rating, the one with more ratings comes first.

```
.\sort-books.cmd condition title -OnShelf
```

That shows only the books you own, best condition first.

```
.\sort-books.cmd language author year:desc -Limit 0 -Csv sorted.csv
```

That saves all 10,000 rows, sorted, to a spreadsheet file.

Sort keys you can use: `title`, `author`, `authorfull`, `year`, `rating`, `popularity`, `language`, `condition`, `status`, `acquired`, `id`. Add `:desc` to any key to reverse it.

Options:

- `-Limit N` shows that many rows (25 by default). `-Limit 0` shows all of them.
- `-OnShelf` keeps only the books you have a physical copy of.
- `-Csv file.csv` saves the results to a file instead of printing them.

If you mistype a key, it tells you which keys are valid.

I tested every one of these, and the CSV export wrote all 10,000 rows.

Sorting details

- **`author`** sorts by last name, so Tolkien files under T. It takes the last word of the first-listed author's name, which can be wrong for a few names, such as "Gabriel García Márquez" filing under Márquez. Use `authorfull` to sort by the name exactly as written.

- `condition` sorts from best to worst (new, good, worn, damaged), not alphabetically. Only the 10 books you have copies of have a condition or status. The other 9,990 have those fields blank, and blanks always go last in any sort. Add `-OnShelf` when you sort by condition or status.
- **Multiple copies:** If you own two copies of a title, it appears twice, once per copy, because each copy has its own condition.

How it works

- **The view:** A new view, `book_catalog`, in `library.db` combines each book with its authors and its copies into one table, so any column can be sorted on. I also added it to `schema.sql`, so a rebuild keeps it.
- **The scripts:** `sort-books.ps1` does the sorting. `sort-books.cmd` exists because Windows blocks PowerShell scripts by default. It runs the script without changing that setting, so always run the `.cmd` version.

YOU · OCT 22, 2026, 7:05 PM

Commit sort-books.ps1, sort-books.cmd, schema.sql and library.db to the assignment branch and push to GitHub.

Bash · Commit sorting tool and database, push

```
cd "C:/Users/johnh/workspace/language-of-computation/week2-assignment" && git branch --show-current && git add sort-books.ps1 sort-books.cmd schema.sql library.db && git commit -q -m "Add book_catalog view and sort-books tool for multi-key sorting

Co-Authored-By: Claude Opus 5.5 <noreply@anthropic.com>" && git push 2>&1 | tail -1 && git status -sb && git log --oneline -1
```

```
assignment
warning: in the working copy of 'schema.sql', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'sort-books.cmd', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'sort-books.ps1', LF will be replaced by CRLF the next time Git touches it
   7ea312c..bbfc45d  assignment -> assignment
## assignment...origin/assignment
bbfc45d Add book_catalog view and sort-books tool for multi-key sorting
```

CLAUDE

I committed `sort-books.ps1`, `sort-books.cmd`, `schema.sql` and `library.db` to the `assignment` branch and pushed them to GitHub. The commit is `bbfc45d`, and your local branch matches GitHub: https://github.com/yuexinghu02/LoC-week2-assignment/tree/assignment

YOU · OCT 22, 2026, 7:05 PM

how do i actually sort though?

CLAUDE

You sort by typing a command in a terminal. It takes three steps.

1. Open a terminal. Use the Terminal panel in this app, or open Windows Terminal or PowerShell from the Start menu.

2. Go to your assignment folder:

```
cd C:\Users\johnh\workspace\language-of-computation\week2-assignment
```

3. Type `.\sort-books.cmd`, then the ways you want to sort, then press Enter:

```
.\sort-books.cmd author
```

That shows the first 25 books, sorted by author last name.

To sort by more than one thing, list them in order of importance:

```
.\sort-books.cmd author year
```

That sorts by author, then by year within each author.

Add `:desc` to reverse a sort, so the biggest or newest comes first:

```
.\sort-books.cmd year:desc
```

Words you can sort by:

Type this	To sort by
`title`	Title, A to Z
`author`	Author's last name
`year`	Publication year
`rating`	Average rating
`popularity`	Number of ratings
`language`	Language
`condition`	Condition, best first (only for books you own)
`status`	Available or on loan (only for books you own)

Extras you can add at the end:

- `-Limit 50` shows 50 rows instead of 25. `-Limit 0` shows all of them.
- `-OnShelf` shows only the 10 books you actually own.
- `-Csv sorted.csv` saves the results to a file you can open in Excel, instead of printing them.

For example, this shows the books you own, best condition first, and then by title:

```
.\sort-books.cmd condition title -OnShelf
```

Each code block above has a Run button, so you can also click that. Commands run from that button start in your assignment folder, so you can skip step 2.

YOU · OCT 22, 2026, 7:29 PM

sort the file by author and give me the first ten rows

PowerShell · Sort catalog by author, show 10 rows

```
Set-Location "C:\Users\johnh\workspace\language-of-computation\week2-assignment"; .\sort-books.cmd author -Limit 10
```

```
book_id                  title                                    authors                  year  rating  ratings
lang   condition  status
-------  ---------------------------------------  ---------------------------------------  ----  ------  -------
-----  ---------  ------
   5816  Why Mosquitoes Buzz in People's Ears     Verna Aardema, Leo Dillon, Diane Dillon  1975    4.01    20639

   2609  Midnight Riot (Peter Grant, #1)          Ben Aaronovitch                          2011    3.92    21519
```

```
eng

   4498  Moon Over Soho (Peter Grant, #2)          Ben Aaronovitch                          2011     4.1    16566
eng

   4922  Whispers Under Ground (Peter Grant, #3)   Ben Aaronovitch                          2012    4.16    15946
eng

   5956  Broken Homes (Peter Grant, #4)            Ben Aaronovitch                          2013    4.16    10958
eng

   6926  Foxglove Summer (Peter Grant, #5)         Ben Aaronovitch                          2014    4.18    12167
eng

   2910  Catch Me If You Can: The True Story of a  Frank W. Abagnale, Stan Redding          1980    4.05    32048
eng
         Real Fake

   8634  حوجن [Ḥawjan]                             Ibraheem Abbas, إبراهيم عباس, Yasser      2013    3.75    10203  ara
                                                   Bahjatt

   3024  Desert Solitaire                          Edward Abbey                             1968    4.22    27723
en-US

   5057  The Monkey Wrench Gang
â€¦ [truncated, 110 more characters]
```

CLAUDE

Here are the first 10 books, sorted by author last name:

#	Author(s)	Title	Year	Rating
1	Verna Aardema, Leo Dillon, Diane Dillon	Why Mosquitoes Buzz in People's Ears	1975	4.01
2	Ben Aaronovitch	Midnight Riot (Peter Grant, #1)	2011	3.92
3	Ben Aaronovitch	Moon Over Soho (Peter Grant, #2)	2011	4.10
4	Ben Aaronovitch	Whispers Under Ground (Peter Grant, #3)	2012	4.16
5	Ben Aaronovitch	Broken Homes (Peter Grant, #4)	2013	4.16
6	Ben Aaronovitch	Foxglove Summer (Peter Grant, #5)	2014	4.18
7	Frank W. Abagnale, Stan Redding	Catch Me If You Can	1980	4.05
8	Ibraheem Abbas, Yasser Bahjatt	حوجن [Ḥawjan]	2013	3.75
9	Edward Abbey	Desert Solitaire	1968	4.22
10	Edward Abbey	The Monkey Wrench Gang	1975	4.08

Only the first-listed author counts for the sort. For example, Aardema puts row 1 first, and the Dillons don't affect it. When one author has several books, they come out in catalog order, not by year. To sort each author's books oldest first, run:

```
.\sort-books.cmd author year -Limit 10
```

None of these 10 books are on your shelves, so the condition and status columns are blank.

YOU · OCT 22, 2026, 7:30 PM

give me the command for git bash

Bash · Test sort-books.cmd from Git Bash

```
cd "C:/Users/johnh/workspace/language-of-computation/week2-assignment" && ./sort-books.cmd author -Limit 3
```

```
book_id                title                                  authors                  year  rating  ratings  lang
condition  status
-------  -----------------------------------  ---------------------------------------  ----  ------  -------  ----
---------  ------
   5816  Why Mosquitoes Buzz in People's Ears  Verna Aardema, Leo Dillon, Diane Dillon  1975    4.01    20639
   2609  Midnight Riot (Peter Grant, #1)       Ben Aaronovitch                          2011    3.92    21519  eng
   4498  Moon Over Soho (Peter Grant, #2)      Ben Aaronovitch                          2011     4.1    16566  eng
```

CLAUDE

In Git Bash, go to the folder first, then use `./` with a forward slash instead of `.\`:

```
cd /c/Users/johnh/workspace/language-of-computation/week2-assignment
```

```
./sort-books.cmd author -Limit 10
```

I tested it in Git Bash and it prints the same results as in PowerShell.

Everything else works the same way. For example:

```
./sort-books.cmd author year -Limit 10
```

```
./sort-books.cmd condition title -OnShelf
```

YOU · OCT 23, 2026, 10:46 AM

output the sorted data in a table in Claude deskop window itself.

Bash · Get first 50 author-sorted rows as JSON

```
cd "C:/Users/johnh/workspace/language-of-computation/week2-assignment" &&
S="/c/Users/johnh/AppData/Local/Microsoft/WinGet/Packages/SQLite.SQLite_Microsoft.Winget.Source_8wekyb3d8bb
we/sqlite3.exe" && "$S" -json library.db "SELECT book_id id, title t, all_authors a, publication_year y,
average_rating r, ratings_count n, language_code l FROM book_catalog ORDER BY author_last_name COLLATE
NOCASE NULLS LAST, primary_author COLLATE NOCASE NULLS LAST, book_id, copy_id LIMIT 50;" | tr -d '\n'
```


```
[{"id":5816,"t":"Why Mosquitoes Buzz in People's Ears","a":"Verna Aardema, Leo Dillon, Diane
Dillon","y":1975,"r":4.01,"n":20639,"l":null},
{"id":2609,"t":"Midnight Riot (Peter Grant, #1)","a":"Ben Aaronovitch","y":2011,"r":3.92,"n":21519,"l":"eng"},
{"id":4498,"t":"Moon Over Soho (Peter Grant, #2)","a":"Ben Aaronovitch","y":2011,"r":4.1,"n":16566,"l":"eng"},
{"id":4922,"t":"Whispers Under Ground (Peter Grant, #3)","a":"Ben
Aaronovitch","y":2012,"r":4.16,"n":15946,"l":"eng"},
{"id":5956,"t":"Broken Homes (Peter Grant, #4)","a":"Ben Aaronovitch","y":2013,"r":4.16,"n":10958,"l":"eng"},
{"id":6926,"t":"Foxglove Summer (Peter Grant, #5)","a":"Ben Aaronovitch","y":2014,"r":4.18,"n":12167,"l":"eng"},
{"id":2910,"t":"Catch Me If You Can: The True Story of a Real Fake","a":"Frank W. Abagnale, Stan
Redding","y":1980,"r":4.05,"n":32048,"l":"eng"},
{"id":8634,"t":"حوجن [Ḥawjan]","a":"Ibraheem Abbas, إبراهيم عباس, Yasser
Bahjatt","y":2013,"r":3.75,"n":10203,"l":"ara"},
{"id":3024,"t":"Desert Solitaire","a":"Edward Abbey","y":1968,"r":4.22,"n":27723,"l":"en-US"},
{"id":5057,"t":"The Monkey Wrench Gang (Monkey Wrench Gang, #1)","a":"Edward
Abbey","y":1975,"r":4.08,"n":15513,"l":"en-US"},
{"id":2931,"t":"Flatland: A Romance of Many Dimensions","a":"Edwin A. Abbott, Banesh
Hoffmann","y":1884,"r":3.81,"n":31377,"l":"en-US"},
```

{"id":6394,"t":"The Fever","a":"Megan Abbott","y":2014,"r":3.12,"n":14775,"l":null},
{"id":6479,"t":"You Will Know Me","a":"Megan Abbott","y":2016,"r":3.45,"n":17002,"l"
â€¦ [truncated, 4946 more characters]

CLAUDE

Here's the catalog sorted by author last name. These are the first 25 of its 10,000 rows:

#	Author(s)	Title	Year	Rating	Ratings	Lang
1	Verna Aardema, Leo Dillon, Diane Dillon	Why Mosquitoes Buzz in People's Ears	1975	4.01	20,639	
2	Ben Aaronovitch	Midnight Riot (Peter Grant, #1)	2011	3.92	21,519	eng
3	Ben Aaronovitch	Moon Over Soho (Peter Grant, #2)	2011	4.10	16,566	eng
4	Ben Aaronovitch	Whispers Under Ground (Peter Grant, #3)	2012	4.16	15,946	eng
5	Ben Aaronovitch	Broken Homes (Peter Grant, #4)	2013	4.16	10,958	eng
6	Ben Aaronovitch	Foxglove Summer (Peter Grant, #5)	2014	4.18	12,167	eng
7	Frank W. Abagnale, Stan Redding	Catch Me If You Can	1980	4.05	32,048	eng
8	Ibraheem Abbas, Yasser Bahjatt	حوجن [Ḥawjan]	2013	3.75	10,203	ara
9	Edward Abbey	Desert Solitaire	1968	4.22	27,723	en-US
10	Edward Abbey	The Monkey Wrench Gang	1975	4.08	15,513	en-US
11	Edwin A. Abbott, Banesh Hoffmann	Flatland: A Romance of Many Dimensions	1884	3.81	31,377	en-US
12	Megan Abbott	The Fever	2014	3.12	14,775	
13	Megan Abbott	You Will Know Me	2016	3.45	17,002	eng
14	Megan Abbott	Dare Me	2012	3.23	15,618	eng
15	Rachel Abbott	Sleep Tight	2014	3.98	10,894	eng
16	Rachel Abbott	Only the Innocent (DCI Tom Douglas #1)	2011	4.03	4,513	eng
17	Rachel Abbott	Stranger Child (DCI Tom Douglas, #4)	2015	4.18	11,856	eng
18	Kōbō Abe, E. Dale Saunders	The Woman in the Dunes	1962	3.91	11,467	eng
19	Joe Abercrombie	The Blade Itself (The First Law, #1)	2006	4.14	97,501	eng
20	Joe Abercrombie	Before They Are Hanged (The First Law #2)	2007	4.27	68,865	eng
21	Joe Abercrombie	Last Argument of Kings (The First Law, #3)	2008	4.26	65,068	eng
22	Joe Abercrombie	Best Served Cold	2009	4.16	36,268	eng
23	Joe Abercrombie	Half a King (Shattered Sea, #1)	2014	3.98	23,911	eng
24	Joe Abercrombie	The Heroes (First Law World, #5)	2011	4.26	24,964	eng
25	Joe Abercrombie	Red Country (First Law World, #6)	2012	4.27	19,926	eng

A blank Lang means the dataset doesn't record a language for that book. Condition and status are left out because none of these books are on your shelves.

When an author has several books, they appear in catalog order, not by year. To show more rows or a different sort, such as author then year, or rating, ask me and I'll show it here the same way.

YOU · OCT 26, 2026, 11:52 AM

make a copy of the transcript of this entire conversation of the week 2 assignment and give it to me in PDF format.

_session_mgmt__export_transcript

```
{"session_id":"self"}
```